WINDOWS SERVER 2019 | ACTIVE DIRECTORY | CYBERSECURITY LAB

Lab 06 — File Server and Permissions

SMB SHARE CONFIGURATION, NTFS ACLS, AND ROLE-BASED ACCESS CONTROLS

Domain Controller: `DC01 (10.1.1.10)`
AD Domain (FQDN): `zroots.local`
NetBIOS Name: `ZROOTS`
Test Client: `WIN10-01 (10.1.1.20)`
Virtual Network: `MyLabNet (10.1.1.0/24)`
UNC Share Path: `\\10.1.1.10\Lab06Data`
Local File Path: `C:\Lab06-FileServer\CompanyData`
Lab Status: FUNCTIONALLY COMPLETED

1. EXECUTIVE SUMMARY

This technical report documents the implementation, security calibration, and role-based access verification of an SMB file share within a Windows Server 2019 Active Directory environment. The primary objective was to deploy a secure directory structure (`Lab06Data`), establish Active Directory Security Groups, configure SMB Share Permissions alongside NTFS Access Control Lists (ACLs) following the principle of least privilege, and validate security boundaries using distinct domain user accounts from a remote Windows 10 workstation (`WIN10-01`).

During initial testing, a permission conflict occurred where share-level Full Access was granted to the Management group, but the underlying NTFS ACL lacked an corresponding Allow entry. This report documents the exact diagnostic workflow, PowerShell ACL remediation, and subsequent re-verification confirming strict access isolation across Read-Only, Modify, and Full Control roles.

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3. LAB OBJECTIVES & SCOPE

- **Directory Provisioning:** Establish a dedicated local file directory tree isolated from existing corporate shares.
- **Role-Based Access Control (RBAC):** Map Active Directory Global Security Groups to access tiers.
- **Dual-Layer Permission Architecture:** Configure SMB Share Access and NTFS ACLs to adhere to least privilege.
- **Access Isolation Verification:** Validate Read-Only, Modify, and Full Control permissions from `WIN10-01`.
- **Troubleshooting Mismatched Permissions:** Resolve Access Denied errors caused by share/NTFS disparity.

4. DIRECTORY ARCHITECTURE

The file server storage was structured on the `C:` drive of `DC01` to represent distinct departmental zones. A dedicated root path was instantiated for this assessment.

```
C:\Lab06-FileServer\CompanyData\
├── HR\                        [Departmental Folder - Reserved]
├── IT\                        [Departmental Folder - Reserved]
└── Lab06-Test.txt             [Primary Role Access Verification File]
```

> **ARCHITECTURAL NOTE & AUDIT FINDING**
>
> During deployment, an administrative path error resulted in the creation of `C:\Users\Administrator\Lab06-FileServer`. To maintain forensic integrity, this unshared directory was left completely untouched and was **not** targeted by the SMB share engine.

5. ACTIVE DIRECTORY GROUPS AND USER MAPPING

Permissions were assigned strictly to Active Directory Domain Security Groups rather than individual accounts. This ensures scalable identity management, auditability, and central access controls.

GROUP / IDENTITY REFERENCE	ASSIGNED ROLE	MEMBER ACCOUNTS	ACCESS PRIVILEGE SCOPE
`ZROOTS\Lab06-File-Read`	Read-Only Role	`student01`	Read & Execute file contents; directory listing.
`ZROOTS\Lab06-File-Modify`	Modify / Contributor	`staff.one`	Read, Write, Append, Delete files within share.
`ZROOTS\Management`	Administrative Control	`ahmed.shah`	Full Control over share, child folders, and files.
`NT AUTHORITY\SYSTEM`	Operating System	N/A (Built-in)	Full Control for background system services.
`BUILTIN\Administrators`	Local Administrators	Domain Admins	Full Administrative Access to filesystem.

6. SMB SHARE CONFIGURATION

The SMB share `Lab06Data` was created in elevated PowerShell on `DC01` targeting the `C:\Lab06-FileServer\CompanyData` path. Explicit share-level access permissions were defined during provision:

```
New-SmbShare -Name "Lab06Data" `
    -Path "C:\Lab06-FileServer\CompanyData" `
    -FullAccess "ZROOTS\Management" `
    -ChangeAccess "ZROOTS\Lab06-File-Modify" `
    -ReadAccess "ZROOTS\Lab06-File-Read"
```

Share permissions were verified using `Get-SmbShareAccess -Name "Lab06Data"`, confirming that `Management` held Full, `Lab06-File-Modify` held Change, and `Lab06-File-Read` held Read permissions at the network abstraction layer.

7. NTFS PERMISSIONS & TARGETED ACL REMEDIATION

The Dual-Layer Permission Model: SMB Access and NTFS ACLs act as dual security gates. When accessing a file over the network, Windows evaluates both layers and applies the *most restrictive combined permission*.

> **IDENTIFIED ROOT CAUSE: DISPARATE PERMISSION LAYERS**
>
> Initial functional testing revealed that user `ahmed.shah` (member of `ZROOTS\Management`) received **Access is Denied** when attempting to append data to `Lab06-Test.txt`. Although the SMB Share granted Full Access, the underlying NTFS filesystem ACL lacked an explicit Allow entry for `ZROOTS\Management`.

To remediate this without disrupting existing pre-existing company shares (e.g., `C:\CompanyData`, which was strictly

8. FUNCTIONAL ACCESS VERIFICATION MATRIX

All tests were executed remotely from domain client `WIN10-01` (10.1.1.20) accessing the UNC network path `\\10.1.1.10\Lab06Data\Lab06-Test.txt` under distinct Active Directory user sessions.

TEST ID	USER / CONTEXT	EXECUTED COMMAND & WORKFLOW	ACTUAL OBSERVED BEHAVIOR	STATUS
Test A	`student01` *(Read Group)*	`whoami` `Get-Content "\\10.1.1.10\Lab06Data\Lab06-Test.txt"`	Verified user identity. Displayed initial file string: *"Lab 06 file access verification."*	PASS

Test B	`staff.one` *(Modify Group)*	`whoami`
		`Add-Content ... "Modify permission test..."`
		`Get-Content ...`
			Command executed successfully. Text appended to file. Repeated execution appended a second identical line.	PASS

Test C	`student01` *(Read Group)*	`Add-Content "\\10.1.1.10\Lab06Data\Lab06-Test.txt" "Denied text"`	PowerShell returned `UnauthorizedAccessException: Access is denied`. Write blocked cleanly.	PASS

Test D	`ahmed.shah` *(Management)*	`whoami`
		`Add-Content ... "Full access test..."`
		`Get-Content ...`
			Executed post-NTFS remediation. Append operation succeeded without error. Full read/write confirmed.	PASS

9. FINAL OBSERVED TEST FILE CONTENT

Following completion of Tests A through D, the final state of `C:\Lab06-FileServer\CompanyData\Lab06-Test.txt` was inspected. The duplicate entry from `staff.one` confirms multiple successful append executions during modify verification.

```
Lab 06 file access verification.
Modify permission test by staff.one.
Modify permission test by staff.one.
Full access test by ahmed.shah.
```

10. TECHNICAL TROUBLESHOOTING ANALYSIS

SYMPTOM / INCIDENT	ROOT CAUSE DIAGNOSTICS	APPLIED RESOLUTION & OPERATIONAL LESSON
Administrator Access Denied	Test was initially initiated as `ZROOTS\Administrator`, which was omitted from SMB Share ACLs.	Role accounts (e.g., `student01`) must be used. Do not grant broad permissions to Administrator simply to bypass ACL design.
Ahmed Access Denied	SMB Share granted Full Access, but underlying NTFS ACL lacked an Allow rule for `ZROOTS\Management`.	Executed targeted PowerShell script to add NTFS Allow rule on `CompanyData` root with inheritance flags.
"Path Does Not Exist" Error	Secondary shell error resulting from UNC path evaluation failure following initial Access Denied condition.	Do not assume file deletion upon path errors. Re-evaluate underlying permission blocks before troubleshooting network storage.
Ahmed Account Lock / Forgotten Password	Account password was unknown during domain login attempt from client workstation.	Reset password via Active Directory Administrative Center. Ensured account was enabled and unlocked. Passwords kept confidential.
Domain Infrastructure Protection	Risk of breaking working AD domain, DNS, or NAT bindings during share troubleshooting.	Avoided domain rejoin, DNS resets, or server rebuilds. Diagnostic efforts remained strictly focused on SMB and NTFS ACL layers.

11. SECURITY LESSONS & BEST PRACTICES

- **Effective Permissions Calculation:** Effective network access is the intersection of SMB Share and NTFS permissions. The standard best practice is granting *Full Control* at the SMB level and managing granular access strictly through NTFS ACLs.
- **Role-Based Access Control (RBAC):** Assigning rights to AD Security Groups rather than individual accounts simplifies administrative overhead, mitigates access drift, and standardizes onboarding/offboarding workflows.
- **Least Privilege & Negative Testing:** Validating that `student01` was explicitly denied write access (Test C) is as crucial as verifying successful write access for authorized accounts.
- **Targeted Remediation Scope:** System fixes must be restricted to the targeted directory tree (`Lab06-FileServer`) without altering pre-existing production shares or modifying domain-wide security settings.

12. EVIDENCE CHECKLIST & VERIFICATION MATRIX

EVIDENCE ITEM	VERIFICATION COMMAND / ARTIFACT DESCRIPTION	VERIFICATION STATUS
SMB Share ACL Verification	`Get-SmbShareAccess -Name "Lab06Data"` output	VERIFIED
NTFS Root ACL Audit	`(Get-Acl "C:\Lab06-FileServer\CompanyData").Access`	VERIFIED
Test A: Read Access Evidence	`student01` successful `Get-Content` output	VERIFIED
Test B: Modify Access Evidence	`staff.one` successful `Add-Content` output	VERIFIED
Test C: Denied Write Evidence	`student01` PowerShell `Access is denied` error	VERIFIED
Test D: Full Control Evidence	`ahmed.shah` post-fix append & read output	VERIFIED
Final File Integrity Audit	Raw monospaced printout of `Lab06-Test.txt`	VERIFIED

13. KNOWLEDGE CHECK & TECHNICAL REVIEW

Q1: Why can a user have Full Access at the SMB layer but still receive "Access is Denied"?
A: Windows evaluates both SMB and NTFS permissions simultaneously. Network access is governed by the most restrictive outcome. If NTFS lacks an Allow rule, SMB Full Access is overridden.

Q2: Which Active Directory group grants read-only access in this lab environment?
A: The `ZROOTS\Lab06-File-Read` group, which contains user account `student01`.

Q3: Why are AD Security Groups preferred over granting direct user permissions?
A: Groups centralize access management. Adding or removing a user from a group automatically updates their access without modifying ACLs across thousands of files.

Q4: What crucial security posture was proven by the denied write attempt in Test C?
A: It proved that the read-only security boundary was successfully enforced and that permissions were properly isolated.

Q5: Why should unrelated shares and paths be left untouched during technical labs?
A: To prevent scope creep, avoid breaking production resources, and maintain clear environment baselines.

> **14. FINAL LAB COMPLETION STATUS**
>
> **Lab 06 — File Server and Permissions: FUNCTIONALLY COMPLETED**
>
> All role-based access tests passed expected parameters. The environment demonstrated robust isolation across Read-Only, Modify, and Management accounts.